🌿 AI FOR SMART AGRICULTURE

Cassava Leaf Disease Classification using EfficientNet & Stratified K-Fold

Team Orbit | KaggleHacX '26

Introduction Overview

What is this project about?

5

Disease Categories classified by EfficientNet from cassava leaf images

Computer Vision Task

Identifies plant diseases from cassava leaf images using deep learning.

Timely Action

Accurate classification helps farmers act quickly to improve crop yield.

EfficientNet Model

Deep learning model based on EfficientNet classifies leaf images accurately.

5 Disease Categories

Images are classified into five distinct cassava disease categories.

Dataset Handling

How the dataset is loaded and split for training

📂 Dataset loaded via PyTorch ImageFolder — class distribution preserved using Stratified K-Fold Cross Validation

📷

ImageFolder Loading

PyTorch ImageFolder organizes images into class-specific folders automatically.

🏆

Stratified K-Fold

Total folds: 3. Each fold maintains the same class proportion for fairness.

🚀

Better Generalization

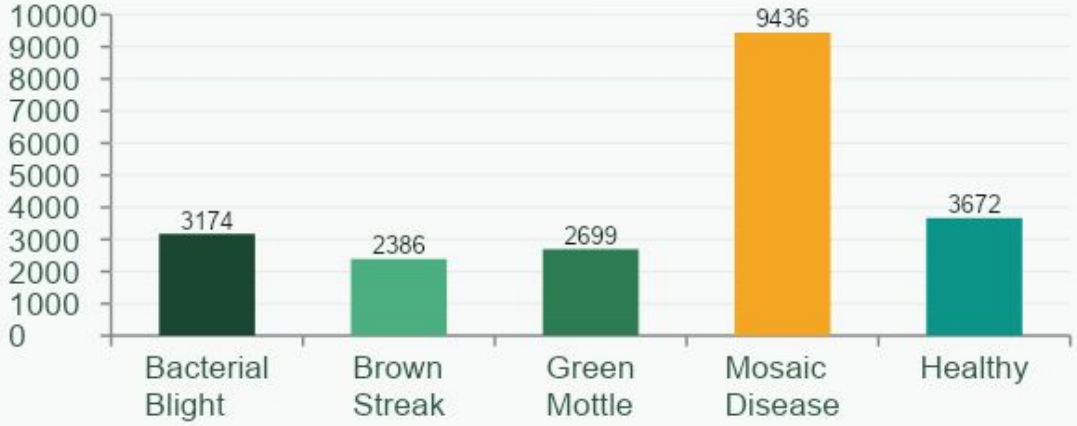
Ensures robust model evaluation across all class categories.

Dataset Overview

Cassava Leaf Disease Recognition — Kaggle Competition Dataset

21,367	5	600×800	ImageFolder
Total Images	Disease Classes	Avg Resolution	Loading Method

Class Distribution



Key Observations

- Mosaic Disease is dominant class (~44%)
- Class imbalance → need stratified split
- ImageFolder preserves class labels
- High-resolution images support fine detail learning
- Visual suggestion: Grid of sample leaf images per class

Data Preprocessing & Augmentation

Transforming raw images into robust training data

🎲 Training Augmentations

- RandomResizedCrop (224×224) — random scale & crop
- Rotation (넓°) — rotational invariance
- Color Jitter — brightness, contrast variations
- Horizontal & Vertical Flip — mirror & orientation diversity
- Normalize (ImageNet mean & std) — stable gradients

✅ Validation Preprocessing

- Resize to 256×256 — consistent input size
- CenterCrop to 224×224 — remove border noise
- Normalize (same ImageNet stats) — aligned with training
- No random transforms — ensures deterministic evaluation

💡 Augmentations simulate real-world variations in lighting, orientation, and scale to improve generalization.

Model Architecture

EfficientNet-B0 pretrained on ImageNet — fine-tuned for cassava classification

Input Layer
224×224×3

MBConv Blocks
B0: 16 layers

Squeeze & Excitation
Channel attention

Dropout (p = 0.4)
Regularization

FC Classifier Layer
5 classes

Compound Scaling

Scales width, depth, and resolution simultaneously — optimal efficiency.

ImageNet Pretrained

Transfer learning from 1M+ images provides strong feature extraction out of the box.

B0: Lightweight & Efficient

B0 is a lightweight baseline with better accuracy and fewer parameters than traditional CNNs.

Dropout Regularization

Added 0.4 dropout before final classifier to reduce overfitting on small datasets.

Final classification layer modified for 5 output disease classes.

Handling Class Imbalance

Techniques to ensure balanced learning across all disease categories

⚖️ Calculate Class Frequencies

Class frequencies are calculated per fold to understand the data distribution.

📊 Inverse Frequency Weights

Inverse frequency weights applied in the loss function to penalize majority classes less.

🛡️ Minority Class Protection

Minority classes are not ignored — model learns from all categories equally.

✅ Balanced Learning

Balanced learning across all categories ensures fair and accurate classification.

⚖️ Weighted Loss Function

Weighted Cross Entropy Loss integrates class weights directly into model training.

Loss Function & Optimization

Weighted Cross Entropy Loss + Adam Optimizer + Cosine Annealing

VAL	TRAIN	TRAIN	TRAIN	TRAIN	Loss Fn
TRAIN	**VAL**	TRAIN	TRAIN	TRAIN	Optimizer
TRAIN	TRAIN	**VAL**	TRAIN	TRAIN	LR: 0.0003
TRAIN	TRAIN	TRAIN	**VAL**	TRAIN	Scheduler
TRAIN	TRAIN	TRAIN	TRAIN	**VAL**	AMP

Applied component Supporting layer

Adam optimizer with lr=0.0003 + Cosine Annealing scheduler smoothly reduces learning rate over all training epochs

Mixed Precision Training

Automatic Mixed Precision (AMP) for faster, memory-efficient GPU training

⚡ AMP Benefits

Reduces Memory Usage

#x2500;

Speeds Up GPU Training

- Uses FP16 for forward pass, FP32 for backward pass
- Cuts memory usage significantly on GPU
- Doubles throughput without sacrificing accuracy
- Enabled via torch.cuda.amp autocast

#xFE0F; AMP Components

autocast

#x2500;

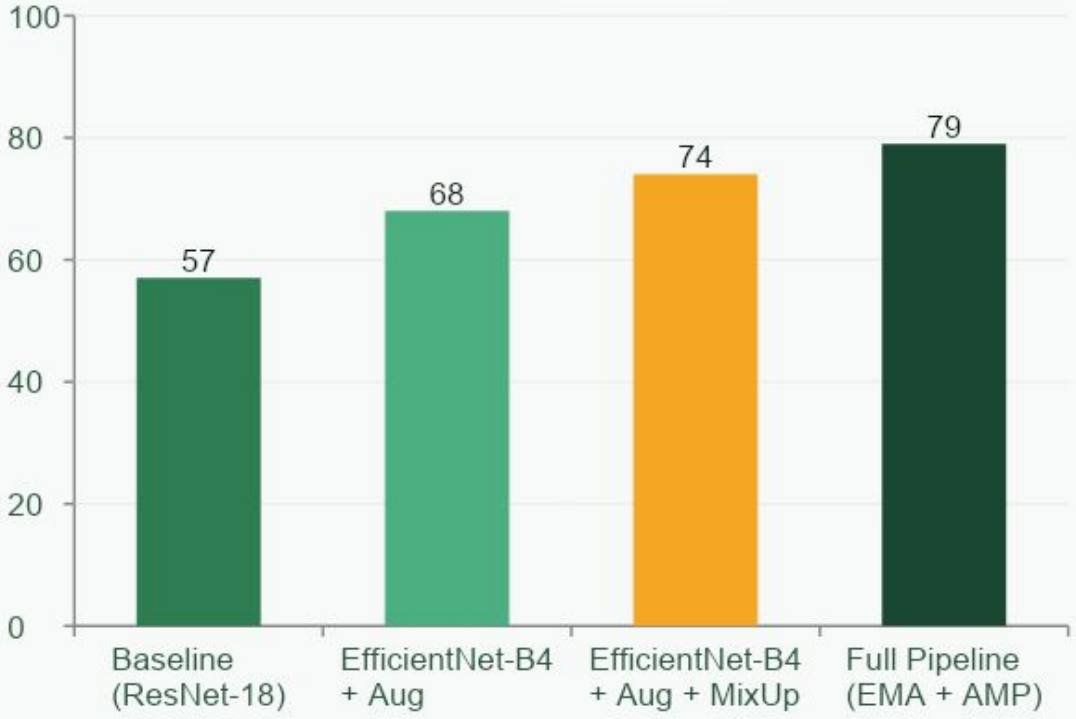
GradScaler

- autocast: runs ops in lower precision (FP16)
- GradScaler: scales gradients to prevent underflow
- Together they preserve numerical stability
- No change to model architecture needed

#x1F4A1; AMP is enabled using torch.cuda.amp — autocast wraps the forward pass and GradScaler handles the backward pass for stability.

Training Strategy

Epoch and fold configuration for full model training



#x1F4C5; Epochs per Fold **3**

Each fold trains for 3 epochs

#x1F504; Total Folds **3**

Stratified 3-Fold cross validation

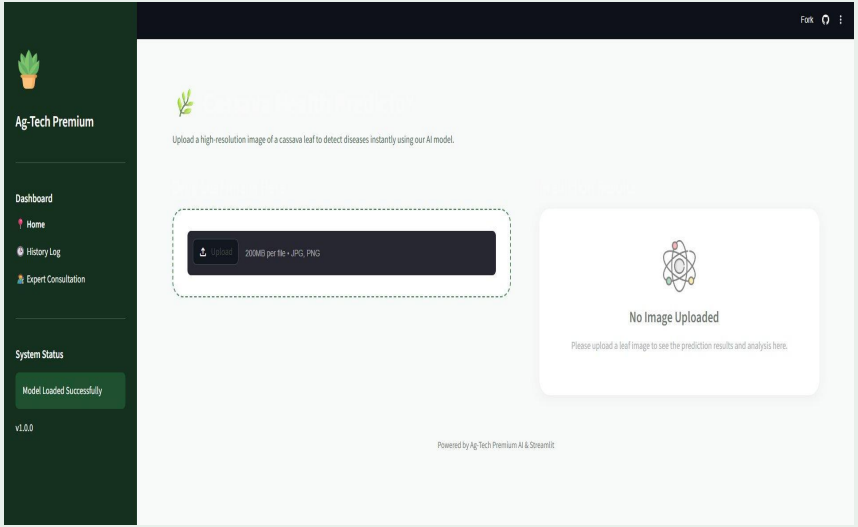
#x1F3AF; Total Cycles **9**

3 folds × 3 epochs = 9 total training cycles

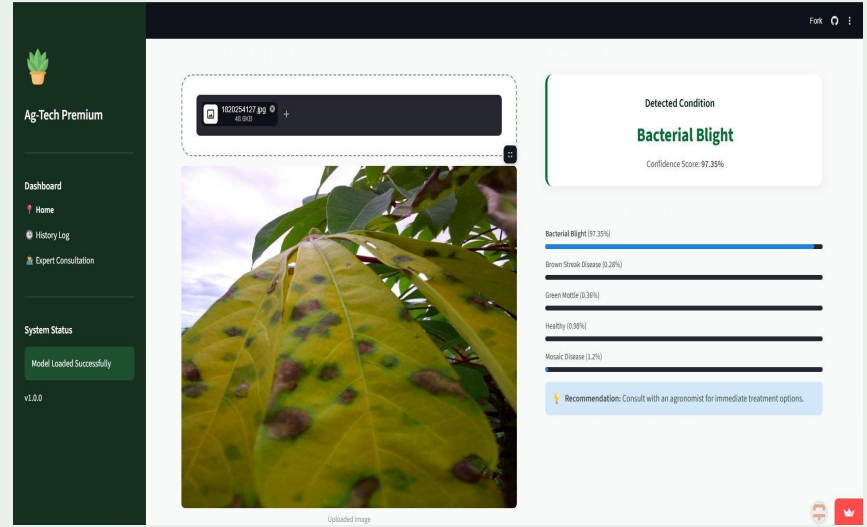
🏁 Per fold: Train on training split → Validate on val split → Compute F1 Score → Save Best Model

Evaluation Metric & Model Saving

How the model is evaluated and the best model is retained



🎯 F1 Score (Weighted) — primary evaluation metric



💾 Best model saved based on highest validation F1 across all folds

#xFE0F; Weighted F1

Handles class imbalance. Weighs precision and recall equally.

#x1F4CA; Imbalance-Aware

Suitable for imbalanced datasets with multiple classes.

#x1F4BE; Global Best Save

Best model saved globally across all folds, not per-fold.

#x2705; Only the Best

Ensures only the best-performing model is retained for inference.

Conclusion

EfficientNet Architecture

EfficientNet-B0 with transfer learning for fast, accurate disease classification.

Stratified K-Fold Validation

3-fold stratified cross-validation ensures robust and fair model evaluation.

Class Imbalance Handling

Weighted loss + inverse frequency weights ensure all classes are learned equally.

Modern Training Techniques

Mixed precision training and cosine LR scheduling further enhance performance.

#x1F331; This approach builds a reliable, scalable cassava disease classification system using EfficientNet, Stratified K-Fold, and class imbalance handling.

Team Orbit

KaggleHacX '26

6-Hour Online

Data Science Challenge

Smart Agri

Deep Learning

EfficientNet-B0